BWI identification utility documentation

Location
/projectnb/turbomac/arozman/share/bwi/bwi_polar_and_event_extraction.py

Explanation
Identifies peaks of DeltaP_blade from events at every azimuth, then links the peaks across adjacent azimuths into continuous BWI arcs. Saves mean event information along those arcs to a csv file: bwi_events.csv

Examples
From CHARM run directory for single rotor:
python bwi_polar_and_event_extraction.py

For multirotor cases:
python bwi_polar_and_event_extraction.py --rotor-id ‘all’

Example bwi events csv file format

arc_id,rotor_id,psi_deg,r_centroid,delp_blade_mean,delp_blade_max,miss_dist_ft,miss_dist_in,vortex_core_ft,vortex_gamma,sweep_perp,vortex_age_rev,n_strikes,primary_blade,vortex_blade
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BWI events output processing documentation

Location
/projectnb/turbomac/arozman/share/bwi/plot_bwi_events.py

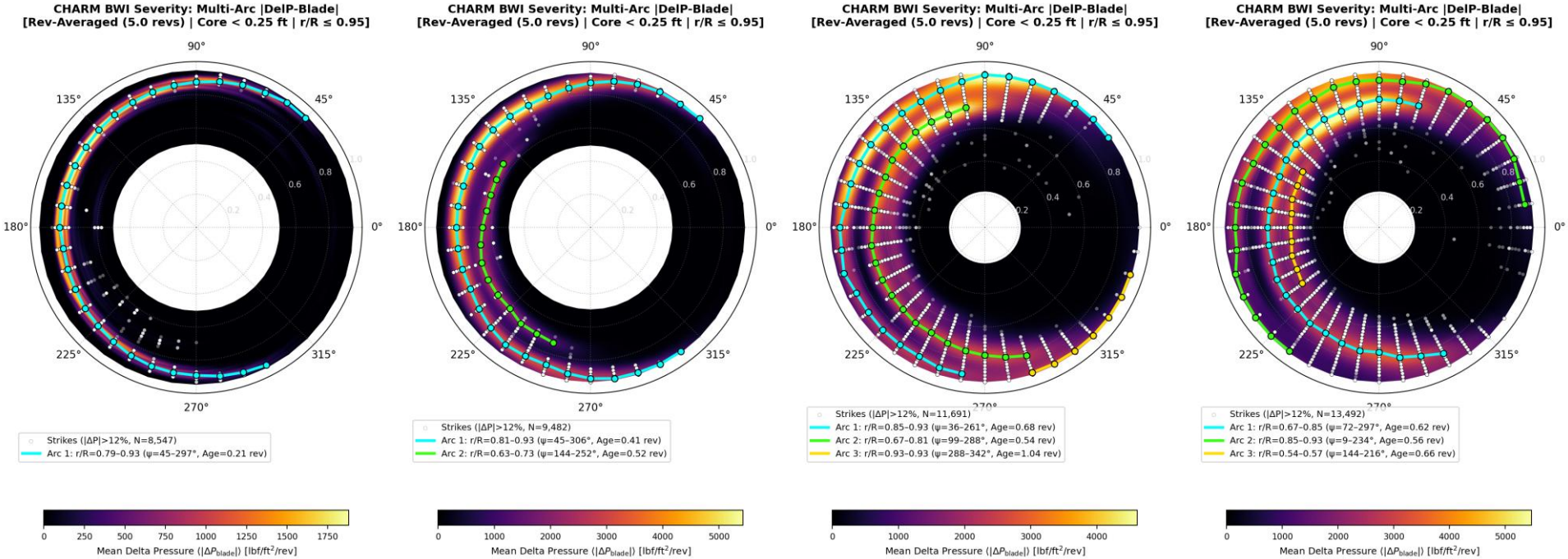
Explanation
Plots some averaged BWI statistics along the arcs extracted by bwi_polar_and_event_extraction.py into bwi_events.csv

Examples
From same directory as bwi_events.csv :
python plot_bwi_events.py

With a specific bwi event csv file:
python plot_bwi_events.py --csv bwi_events_rotor1.csv

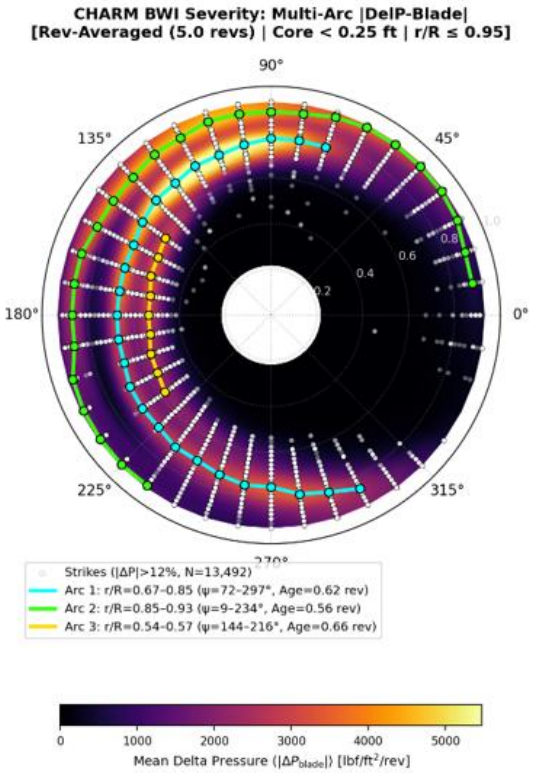
BWI Recognition: Grouping into Event Arcs

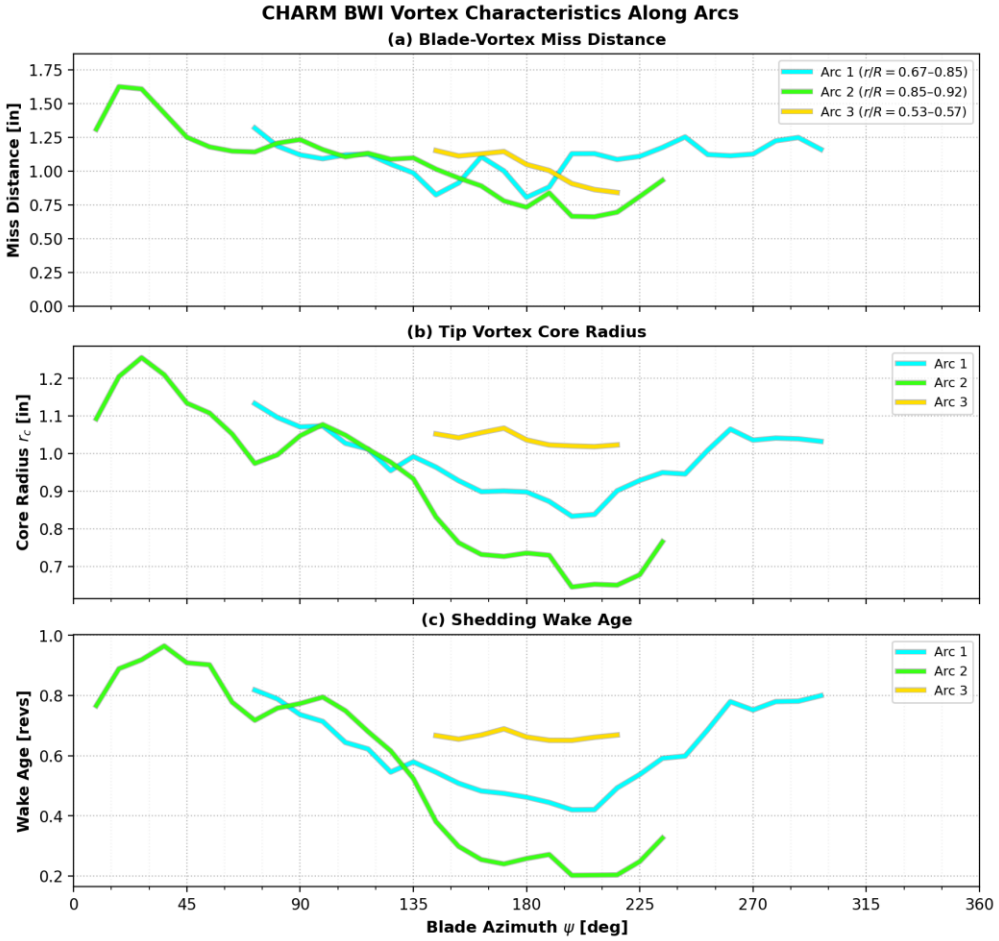
Tilt angle (°): -20 -10 10 20



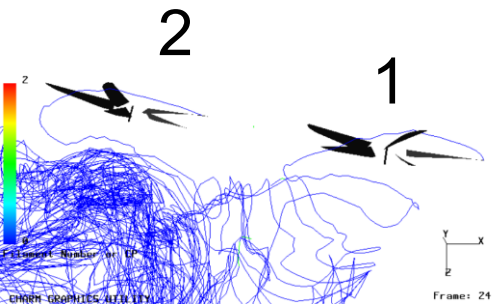
Ascent → Descent

plot_bwi_events.py visualization





Dual-Prop BWI information



- This is the case where front rotor has the vortex passing *over* advancing blade, rear has vortex passing *under*.

